

SWASTHAM.AI · PROOF-OF-CONCEPT REVIEW

One Answer, Two Languages

English–Hindi parity for health answers: the problem, the solution, the bugs we hit, and what carries into production.

25 September 2026

English question

Hindi question

ONE
canonical
answer

same evidence · same facts · same cautions

At a glance

1

canonical English answer per question, reused for every language

15

English/Hindi evaluation pairs, 6 of them safety-critical

25

defects found and fixed across 3 rounds

32

offline unit tests (up from 14), all passing

PROBLEM

The same health question asked in English and in Hindi produced two different answers, and the Hindi one sometimes dropped safety cautions.

SOLUTION

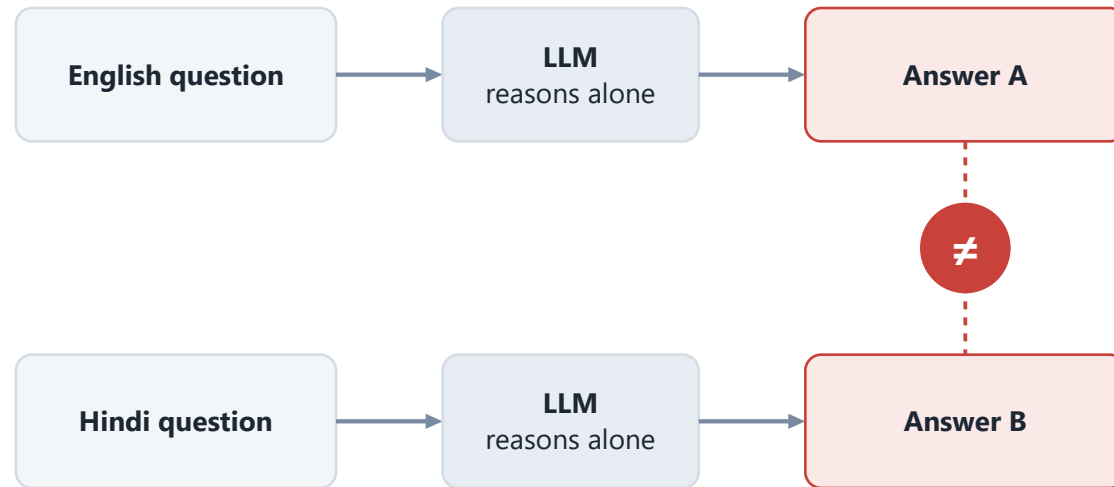
Hindi never reasons on its own: one grounded English answer is generated, validated, then faithfully translated and checked for parity.

STATUS

All UI samples pass parity. Hardened in 3 rounds. Open item: keyword checks need meaning-level replacements before production.

Two languages, two independent answers

BEFORE · each language generated its own answer

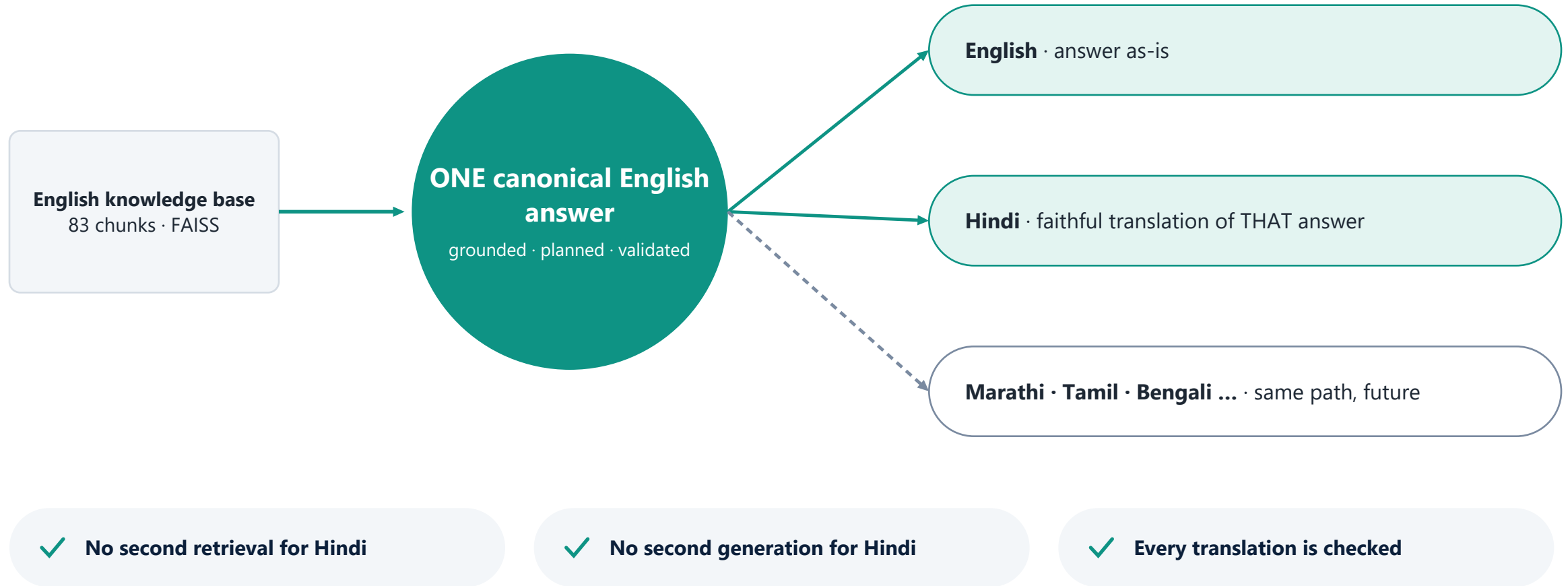


Retrieval, reasoning and wording all drift between the two runs.

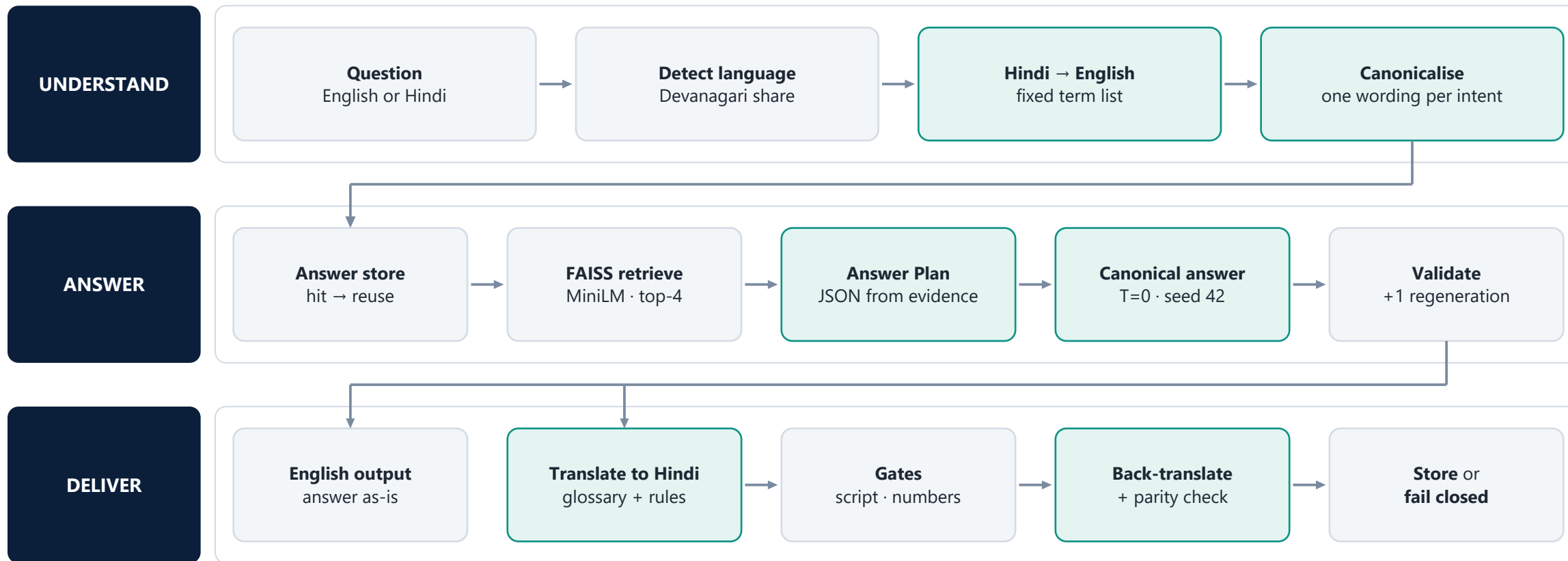
- 1 Different facts**
Numbers, ranges and cut-offs differ between languages.
- 2 Less detail**
The Hindi answer is shorter and skips points the English one makes.
- 3 Safety caution dropped**
e.g. "adjust your insulin dose with your doctor" missing in Hindi.

For a health product, two different answers to one question is a patient-safety risk, not a UX issue.

Hindi never reasons on its own



End-to-end answer path



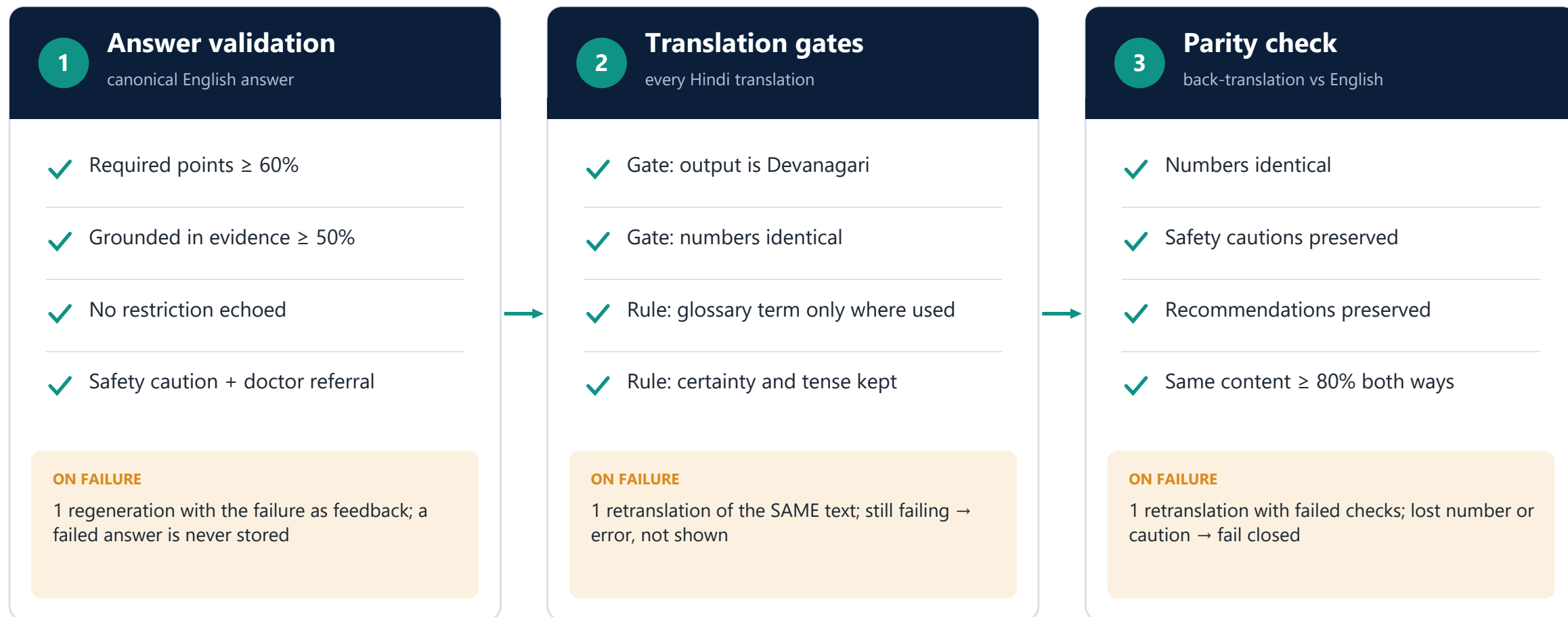
 LLM call (Groq, gpt-oss-120b, temperature 0)

 Local and deterministic (no LLM)

6 of 14 steps call the LLM

An English and a Hindi user asking the same thing converge at **Canonicalise** → same store entry → same chunks → same answer.

Three layers of checks, each with one retry



Fail closed: a Hindi answer that lost a number or a safety caution is never shown to the user.

Six problems, six mechanisms

1 **PROBLEM**
Divergent answers

MECHANISM
Single canonical English answer

GUARANTEE
Same facts in both languages

2 **PROBLEM**
Same intent, different evidence

MECHANISM
Canonicalisation + answer store

GUARANTEE
Same question → same chunks

3 **PROBLEM**
Unfaithful translation

MECHANISM
Glossary, strict prompt, hard gates

GUARANTEE
Script and numbers enforced

4 **PROBLEM**
Lost safety cautions

MECHANISM
Validator + parity caution check

GUARANTEE
Fail closed, never shown

5 **PROBLEM**
Non-determinism

MECHANISM
T=0, seed, stored checked answers

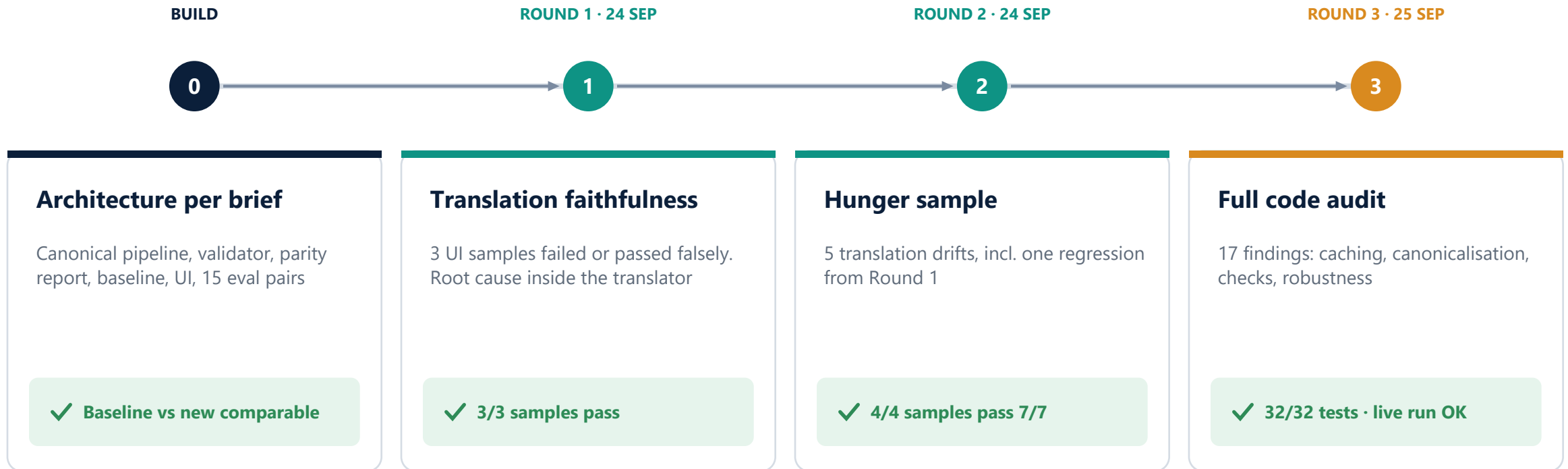
GUARANTEE
Repeat question → same text

6 **PROBLEM**
Checks that pass silently

MECHANISM
Sanity conditions on check inputs

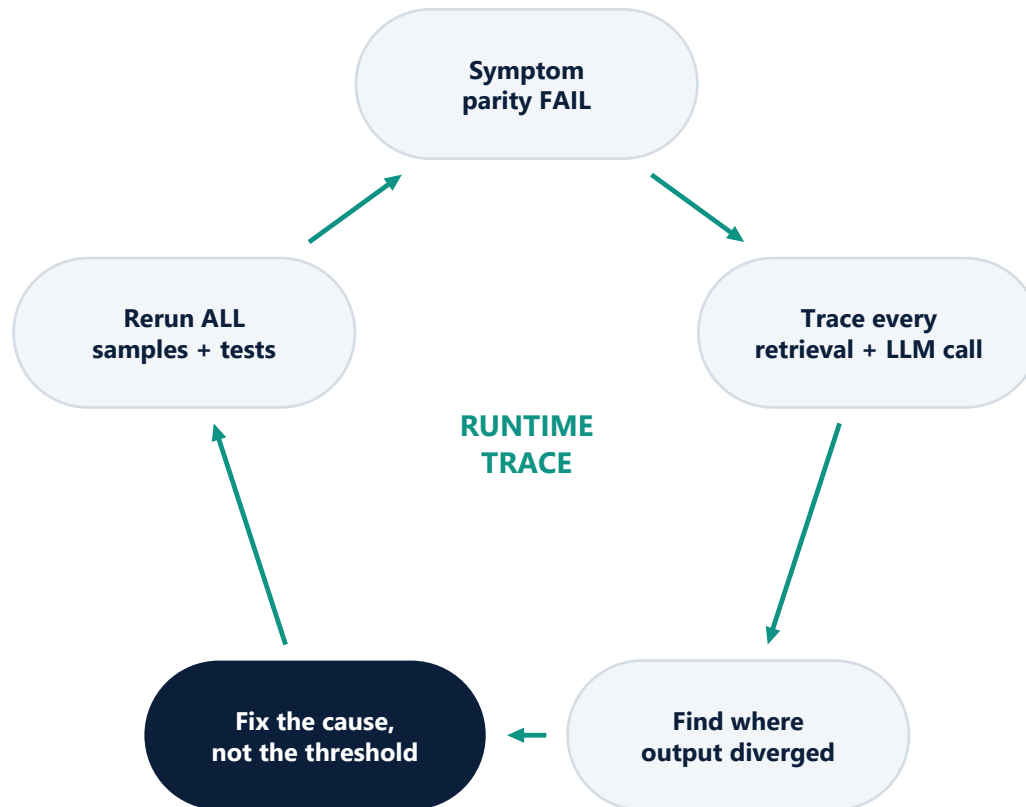
GUARANTEE
A pass actually means something

Build, then three rounds of fixes



Each round: **trace the runtime** → **find the divergence** → **fix the cause** → **rerun every sample**

Trace the runtime before changing code



WHAT THE TRACE PROVED IN ROUND 1

- ✓ Exact canonical string was sent to the translator
- ✓ Exactly 1 retrieval; no second RAG pass for Hindi
- ✓ 0 independent Hindi generations
- ✓ No post-processing after translation
- ✗ The Hindi text itself had changed the content

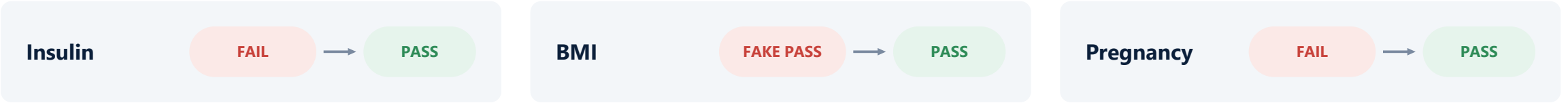
Insight

The architecture was correct. The fault was the translator. Minutes of tracing answered what reading the code could not.

The translator changed the content

SAMPLE	WHAT THE TRANSLATOR DID	EFFECT ON PARITY	FIX
BMI cut-offs	Returned the English text untouched, no Hindi at all	False PASS: English compared with English	→ Devanagari gate on every translation
Pregnancy	"obstetrician" became "obstetric specialist"	Doctor referral lost → Safety FAIL	→ Glossary entries for referral roles
Insulin	"clinician" collapsed to "doctor"; "almost certainly" weakened	Recommendation FAIL; certainty softened	→ Keep-certainty + keep-role prompt rules

RESULT · same stored English answers, so the translator got identical input before and after



Five subtle drifts, one of them self-inflicted

1	Tense changed	"take medicines" → "are taking"	Recommendation verb lost	→	Keep-tense rule; literal back-translation verbs
2	Term added	"diabetes" → "type 2 diabetes"	Content not in the source	→	Glossary term only where the English uses it
3	Role changed	"healthcare professional" → "healthcare team"	Regression from Round 1 glossary	→	Separate entries per role
4	Heading skipped	Heading left in English	Partial translation	→	Translate headings and bold labels
5	Echo on retry	BMI answer echoed in English, twice (fixed seed)	Gate refused it, correctly	→	Repeat instruction AFTER the source text

Lesson: a glossary fix in Round 1 caused drift #3. Every change needs a rerun of every sample.

✓ 4 samples · 7/7 parity checks each

17 findings, confirmed before fixing

FINDINGS BY SEVERITY

High



wrong or unchecked answers reached users

Medium



reliability, determinism, false positives

Low



config, docs, input handling, tests

Findings were **confirmed** on real data (stored answers, canonicaliser output) before any code changed.

THE FIVE HIGH-SEVERITY FINDINGS

Failed answers cached forever

2 safety-critical insulin answers stored despite failing validation

English never truly canonicalised

Only lower-cased: "What's BMI" and "What is BMI?" got different answers

Hindi rules collapsed intents

"Which diseases does obesity cause?" became "What is obesity?"

Truncated LLM output accepted

Cut-off answers could be validated, stored and translated

3 of 7 parity checks always passed

Two compared a trace with its own copy; one was a duplicate

Before and after the audit

AREA	BEFORE	AFTER
Answers that failed validation	✗ Cached and reused (2 insulin answers)	✓ Never stored; rebuilt next time
Stale answers after a prompt change	✗ Manual delete of the store	✓ Fingerprint discards them automatically
English canonicalisation	✗ Lower-casing only	✓ LLM, once per wording, stored
Hindi rule-based canonicaliser	✗ Collapsed different intents	✓ Removed
Cut-off or empty LLM output	✗ Used silently	✓ Rejected with an error
Parity checks that always pass	✗ 3 of 7	✓ 0
Parity on the Ask tab	✗ Not run	✓ Checked, 1 retry, fail closed
Safety keyword matching	✗ Substrings ("kid" in kidney)	✓ Word-start matching
Server and store	✗ Single-threaded, plain writes	✓ Threaded, locked, atomic writes
LLM retries	✗ Rate limits only	✓ Rate limits, 5xx, timeouts
Offline unit tests	✗ 14	✓ 32

Stack and effort

SYSTEM LAYERS

Web UI

Ask tab · Parity Compare · pipeline trace

HTTP API

Python standard library, threaded

Pipeline

canonicalise · store · plan · answer · validate · translate · parity

Models

Groq gpt-oss-120b (LLM) · MiniLM-L6 (embeddings) · FAISS (index)

Knowledge base

English obesity / diabetes document · 83 chunks

~2,100

lines of Python across 8 modules

~1,200

lines of web UI

6–10

LLM calls per new comparison

3

fix rounds in 2 days

15

EN/HI evaluation pairs

32

offline tests, no API key needed

Nine lessons

01

Generate once, translate everywhere

One reasoning path scales to every language.

02

Validate translations like model output

A correct architecture still let the translator drift.

03

Fail closed on safety

Never show a Hindi answer that lost a caution.

04

A passing check can hide a bug

English vs English scored 100%.

05

The glossary is a clinical asset

One term decides whether advice survives.

06

Normalise questions before retrieval

Same intent must hit the same evidence.

07

Trace the runtime first

Logs answered what code reading could not.

08

Fix causes, not thresholds

A lower bar would have hidden a lost referral.

09

Determinism is a setting, not a guarantee

Stored, checked answers give consistency.

Roadmap for the real system



Known limitations, stated honestly

HIGH

Keyword checks give false positives and negatives

The insulin two-meal answer fails restriction_compliance for correctly saying “not a cure”, so it is regenerated on every request.

MEDIUM

LLM output is not fully deterministic

Temperature 0 and a seed reduce variation; stored checked answers remove it for repeat questions.

MEDIUM

Canonicalisation can split one intent

Two wordings may canonicalise differently; Parity Compare now shows this explicitly.

MEDIUM

Glossary coverage is document-specific

New roles, drugs or topics need entries in both directions.

LOW

Free-tier rate limits

Heavy testing waits on Groq limits; evaluation needs a paid tier.

STATUS

Full 15-pair evaluation not rerun since the audit

Unit tests and a live end-to-end run pass; the evaluation is the next step.

THE TAKEAWAY

Consistency across languages is architecture plus verification.

1

One canonical answer

Hindi never reasons on its own

2

Checked translation

Gates, back-translation, parity, one retry

3

Fail closed

Unsafe Hindi is never shown